PRAKHAR SHARMA

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[LinkedIn] — [GitHub] — [Portfolio]

SUMMARY

CS (AI) student focused on applied AI and full-stack engineering. Built and deployed LLM systems and a real-time multiplayer game, tested with pytest and GitHub Actions CI. Seeking AI/ML and full-stack internships.

EDUCATION

Technocrats Institute of Technology — Bhopal, India
Bachelor of Technology in Computer Science (Artificial Intelligence) — **2023 – 2027**
The Radiant School — Senior Secondary Education (Class XII) — Gwalior, India — **2022**

TECHNICAL SKILLS

- **Languages:** Python, C++, JavaScript, GDScript
- **AI/ML:** Generative AI (LLM apps), LLM integration (Groq), RAG, FastEmbed (ONNX), Prompt Engineering, Prompt-Injection Defence, LLM Evaluation, Speech-to-Text (Web Speech API)
- **Backend:** FastAPI, REST APIs, WebSockets, Server-Sent Events, JWT Auth, slowapi (rate limiting)
- **Frontend & Game Dev:** React, Next.js, Tailwind CSS, Godot Engine
- **Databases:** MongoDB (Motor), Supabase (PostgreSQL, Auth, Realtime), Qdrant (vector DB)
- **Cloud & DevOps:** Docker, Git, GitHub Actions (CI), Pytest, Playwright, Sentry, Vercel, Render

PROJECTS

Qualif-AI — AI Interview Simulator — *FastAPI, React, MongoDB, Groq* — 2026 — [Live] — [GitHub]

- Built a voice-first, 4-phase interview flow (Introduction, Resume, Technical, Behavioural) with a skill graph of 42 skills across 7 categories that tracks mastery over each user's last 5 scores.
- Engineered hybrid scoring that blends LLM judgement (72–80%) with deterministic checks (20–28%), plus a 5-layer prompt-injection defence (regex sanitiser, delimited input, prompt rule, score floor, LLM classifier).
- Wrote 277 backend tests and 4 Playwright E2E specs (69% coverage on application code), run in a GitHub Actions CI pipeline (ruff, mypy, pytest, Playwright).

agAInst — AI Debate Platform — *Next.js, FastAPI, Supabase, Groq (gpt-oss-120b)* — 2026 — [Live] — [GitHub]

- Built a debate arena where users argue against an AI debater and an independent AI judge scores logic, evidence, rebuttal and clarity; designed 5 LLM prompts with JSON validation and retry; migrated from Llama 3.3 70B to gpt-oss-120b after Groq's deprecation.
- Implemented online PvP with Supabase Realtime (postgres_changes), auth with cloud sync, a global leaderboard, and a pg_cron job that marks stale rooms as abandoned.
- Added rate limiting (slowapi, HTTP 429), prompt-injection sanitising of player input, and 35 s client timeouts.
- Wrote 44 pytest tests (97% backend coverage) with a mocked LLM and a GitHub Actions CI workflow; stress-tested the live API, traced 500 errors to Groq's shared token limit, and fixed them with caching and a graceful fallback.

DocuMind — RAG Document Q&A — *FastAPI, Qdrant, Groq, Next.js, Docker* — 2026 — [Live] — [GitHub]

- Built a RAG pipeline: PyMuPDF ingestion, overlapping chunking (512 characters, 64 overlap), MiniLM embeddings, and per-document filtered vector search using a Qdrant payload index.
- Streamed answers over Server-Sent Events with source citations sent before the first token; added an evaluation endpoint and dashboard scoring context recall, faithfulness and relevance; containerised the backend with Docker.
- Fixed a 512 MB out-of-memory crash on Render by replacing PyTorch with an ONNX runtime (FastEmbed), and migrated from retired Groq Llama models to gpt-oss.

Mafia Web — Real-Time Multiplayer Game — *FastAPI WebSockets, Next.js* — 2026 — [Live]

- Built a Town of Salem-style game joined via room codes: FastAPI WebSocket backend with server-authoritative state, Next.js frontend, and role abilities (Godfather, Doctor, Detective, Villager) for 5+ players.

CodeQuest: Syntax Scrolls — *Godot Engine, GDScript, Python* — MVP, in development — [GitHub]

- Built a 2D RPG where players solve puzzles by writing executable Python, with a custom in-game evaluation engine that checks outputs and gives targeted syntax feedback.
- Designed modular, data-driven subsystems (movement, dialogue, quests) so new puzzles can be added without changing core engine code.

INTERNSHIP

AWS & DevOps Intern (Virtual) — EWB Edutech Pvt. Ltd. (via National Internship Portal) Oct 2026 – Present

- Completing a mentor-led AWS and DevOps programme with weekly assessments, assigned tasks and a final mini project.

CERTIFICATIONS

Google AI Essentials (2026) — Intro to Generative AI (2025) — Intro to Responsible AI (2025)